1 Seed Initialization

Bilingual Niche Entities → Balanced Seed Bank

Bilingual Entity KGs

Wiki-EN/ZH

Filtering

low in-/out-degree entities

Random Walks × N

Topology Potential

Domain-aware Balancing

Prior Domains + incremental label updates

Keep sampling counts approximately balanced across first-level domains

Sci. Hist. Cult. ...

2 Web-enhanced Agentic Knowledge Expansion

Iterative Web Search → Topology-rich Global Knowledge Graph

SEED

Query Agent

propose queries beyond parametric knowledge

Multi-source Web Pages

Fact Extraction

retain uncommon relations & evidence

GLOBAL KG

Iterative Expansion

detect missing-link local regions → ask targeted follow-up queries → disambiguate + aggregate across pages

repeat until the graph becomes deeper, wider, and locally topology-diverse

3 Knowledge Sampling with Difficulty Ranking

Constrained Subgraphs → High-difficulty Candidates

Massive constrained sampling

nodes depth width

frontier expansion

Difficulty Score D(G)

- avg. entity in/out degree
- minimum depth
- minimum width
- CV of ⟨query ID, document ID⟩

rank all sampled subgraphs

retain top-k high-D

4 QA Generation with Complexity Scaling

Strong-LLM Synthesis, Iterative Escalation, and Trajectory-conditioned Difficulty Floors

Initial Hard-QA Synthesis

High-D Subgraph

Strong LLM

Initial Triplets

⟨question, answer, factual reasoning chain⟩

Multiple Candidates

Object Fuzzing | Multi-hop Facts | Tool-solvable Steps

Entities, relations, and values are deliberately obscured; every hop maps to deep search or fuzzy inversion.

Iterative Complexity Scaling Engine

Deep Search | Wide Search

1. **Reasoning-chain Expansion**
   Extend factual chains over the sampled subgraph; add steps requiring deeper and broader search.
2. **Trajectory-conditioned Escalation**
   Synthesize rollouts → stratify by Pass@K → sample long-horizon hard trajectories as generation context.
3. **Raise the Difficulty Level**
   Recycle the selected context into the next synthesis round; iterate until complexity stabilizes.

Final QA Outputs

Multi-Style

Benchmark-style

complex multi-hop descriptive question + concise grounded answer

Conversational

natural question + paragraph-length answer (wide search)

Factual Reasoning Chain Retained